

6.7980: Topics in Multiagent Learning

Lecture 6 – Prof. Daskalakis
Fall 2026



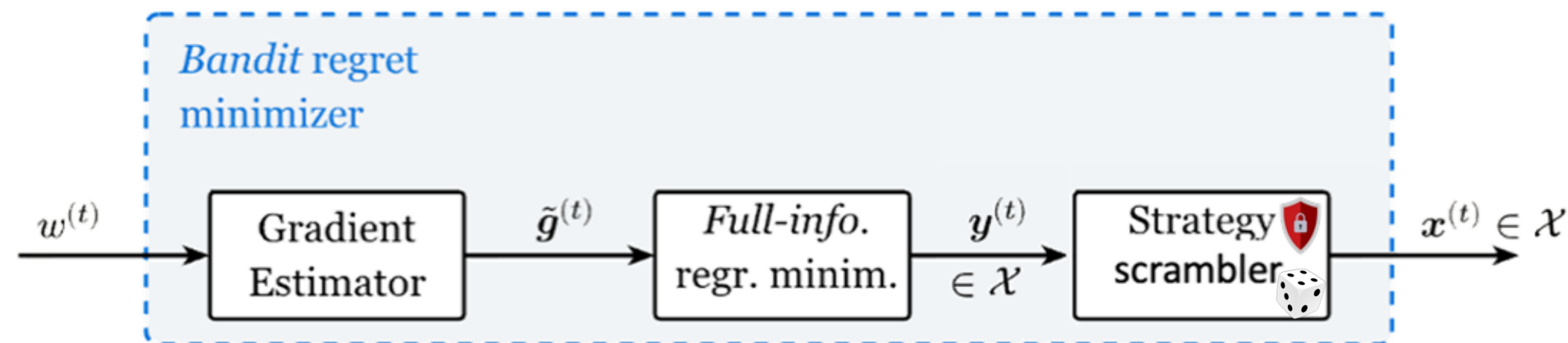
Refresher: Full-Info Online Learning Setting

- For $t = 1, \dots, T$:
 - Learner chooses $\mathbf{x}^{(t)} \in \mathcal{X} \subset \mathbb{R}^n$
 - Environment chooses $\mathbf{g}^{(t)} \in \mathbb{R}^n$
 - Learner receives utility $w^{(t)} = \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle$ and observes utility function $u^{(t)}(\cdot) = \langle \mathbf{g}^{(t)}, \cdot \rangle$
- How does the environment choose $\mathbf{g}^{(t)}$?
 - We want to remain agnostic to this, because we want whatever we prove to be super robust
 - So, we will assume that the $\mathbf{g}^{(t)}$ are chosen by an adversary
 - In games, $\mathbf{g}^{(t)}$ result from the choices made by the other players at step t
- Learner's External Regret: $\text{Reg}^{(T)} = \max_{\hat{\mathbf{x}} \in \mathcal{X}} \left\{ \sum_{t=1}^T (\langle \mathbf{g}^{(t)}, \hat{\mathbf{x}} \rangle - \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle) \right\}$
- When $\mathcal{X} = \Delta(A)$, $\mathbf{x}^{(t)}$ can be viewed as a probability distribution over actions in A
 - External Regret can equivalently be written as:
$$\text{Reg}^{(T)} = \max_{a \in A} \left\{ \sum_{t=1}^T (\langle \mathbf{g}^{(t)}, \mathbf{e}_a \rangle - \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle) \right\}$$
$$= \max_{a \in A} \left\{ \sum_{t=1}^T \left(g_a^{(t)} - \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle \right) \right\}$$

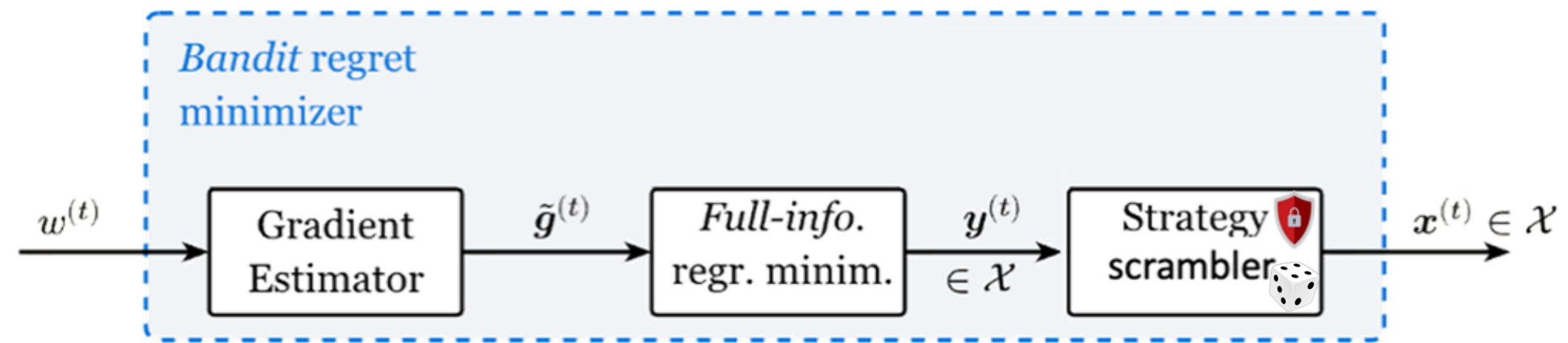
Last time...

Online Learning with Bandit Feedback

- For $t = 1, \dots, T$:
 - Learner chooses $\mathbf{x}^{(t)} \in \mathcal{X} \subset \mathbb{R}^n$
 - Environment chooses $\mathbf{g}^{(t)} \in \mathbb{R}^n$
 - Learner receives utility $w^{(t)} = \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle$ but does not observe utility function $u^{(t)}(\cdot) = \langle \mathbf{g}^{(t)}, \cdot \rangle$
- This poses a problem with running algos that use full information feedback, i.e. not just $w^{(t)}$ but the whole $u^{(t)}$
 - The algorithms we saw for $\mathcal{X} = \Delta(A)$, all have this property!
- General algorithmic architecture to tackle this issue:

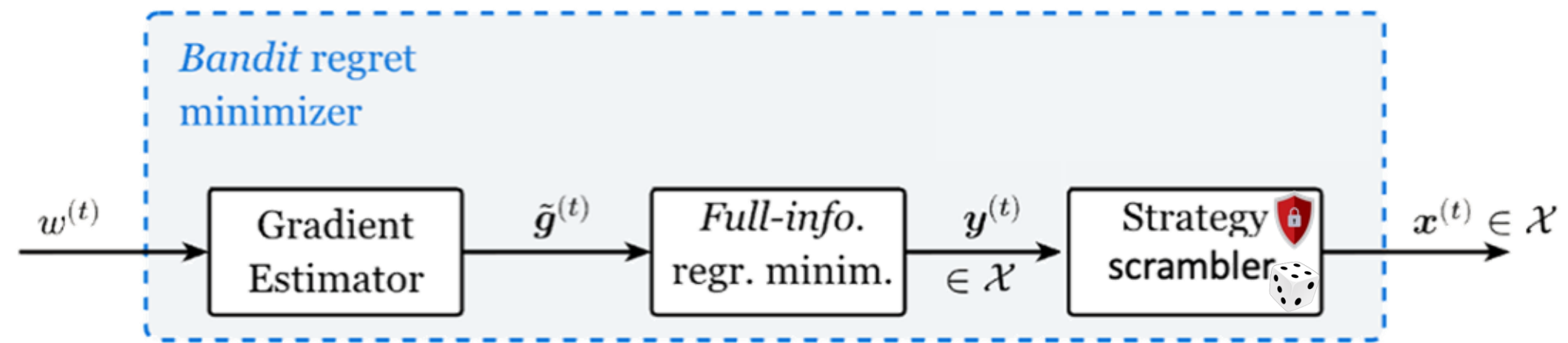


Let's execute on this approach for $\mathcal{X} = \Delta(A)$



- We will use:
 - for “Full-info regret minimizer,” the MWU algorithm
 - for “Strategy scrambler,” the algorithm that samples $\mathbf{y}^{(t)}$
 - i.e. $\mathbf{x}^{(t)} = \mathbf{e}_{a^{(t)}}$, where $a^{(t)} = \{a, \text{with probability } y^{(t)}(a)\}$
- So, at every round t :
 - Learner plays some deterministic action $a^{(t)}$ and observes only that coordinate of $\mathbf{g}^{(t)}$, i.e. she observes $w^{(t)} = g^{(t)}(a^{(t)})$
 - But adversary can only predict $\mathbf{y}^{(t)}$ and not the sample realization $a^{(t)}$
- **Question:** Given $w^{(t)}$ is there an unbiased estimator of $\mathbf{g}^{(t)}$?
- **Answer:** Yes, the *importance sampling estimator* $\tilde{\mathbf{g}}^{(t)} = \left(\frac{w^{(t)}}{y^{(t)}(a^{(t)})} \right) \mathbf{e}_{a^{(t)}}$
- **Theorem:** Assume $y^{(t)}(a) > 0$, for every action a . Then, the importance sampling estimator is unbiased, that is, $\mathbb{E}[\tilde{\mathbf{g}}^{(t)}] = \mathbf{g}^{(t)}$.
- **Proof:** $\mathbb{E}[\tilde{\mathbf{g}}^{(t)}] = \mathbb{E}_{a^{(t)} \sim y^{(t)}} \left[\left(\frac{w^{(t)}}{y^{(t)}(a^{(t)})} \right) \mathbf{e}_{a^{(t)}} \right] = \mathbb{E}_{a^{(t)} \sim y^{(t)}} \left[\left(\frac{g^{(t)}(a^{(t)})}{y^{(t)}(a^{(t)})} \right) \mathbf{e}_{a^{(t)}} \right] = \sum_a y^{(t)}(a) \left(\frac{g^{(t)}(a)}{y^{(t)}(a)} \right) \mathbf{e}_a = \sum_a g^{(t)}(a) \mathbf{e}_a = \mathbf{g}^{(t)}$.

Let's execute on this approach for $\mathcal{X} = \Delta(A)$



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- **Theorem:** Assume $y^{(t)}(a) > 0$, for every action a . Then, the importance sampling estimator is unbiased, that is, $\mathbb{E}[\tilde{\mathbf{g}}^{(t)}] = \mathbf{g}^{(t)}$.
- Importance sampling estimator could explode upwards.
 - Here is a variant that doesn't: $\tilde{\mathbf{g}}^{(t)} = \mathbf{1} + \left(\frac{w^{(t)} - 1}{y^{(t)}(a^{(t)})} \right) \mathbf{e}_{a^{(t)}}$ - still $\mathbb{E}[\tilde{\mathbf{g}}^{(t)}] = \mathbf{g}^{(t)}$ but now $\mathbf{g}^{(t)} \leq \mathbf{1}$

The Exp3 Algorithm

EXP3 Algorithm

Initialize: $y^{(1)}(a) = \frac{1}{|A|}, \forall a$

$\forall t \geq 1$: sample: $a^{(t)} \sim y^{(t)}$, observe: $g^{(t)}(a^{(t)})$

estimate: $\tilde{g}_t(a) = \begin{cases} 1, & \text{if } a \neq a^{(t)} \\ 1 + \frac{g^{(t)}(a^{(t)}) - 1}{y^{(t)}(a^{(t)})}, & \text{if } a = a^{(t)}, \forall a \end{cases}$

update: $y^{(t+1)}(a) = \frac{y^{(t)}(a) \cdot \exp(\eta \cdot \tilde{g}_t(a))}{\sum_{a'} y^{(t)}(a') \cdot \exp(\eta \cdot \tilde{g}_t(a'))}, \forall a$

Theorem: Suppose $g^{(t)} \in [0,1]^A$ and . Then EXP3 incurs *pseudo-regret*

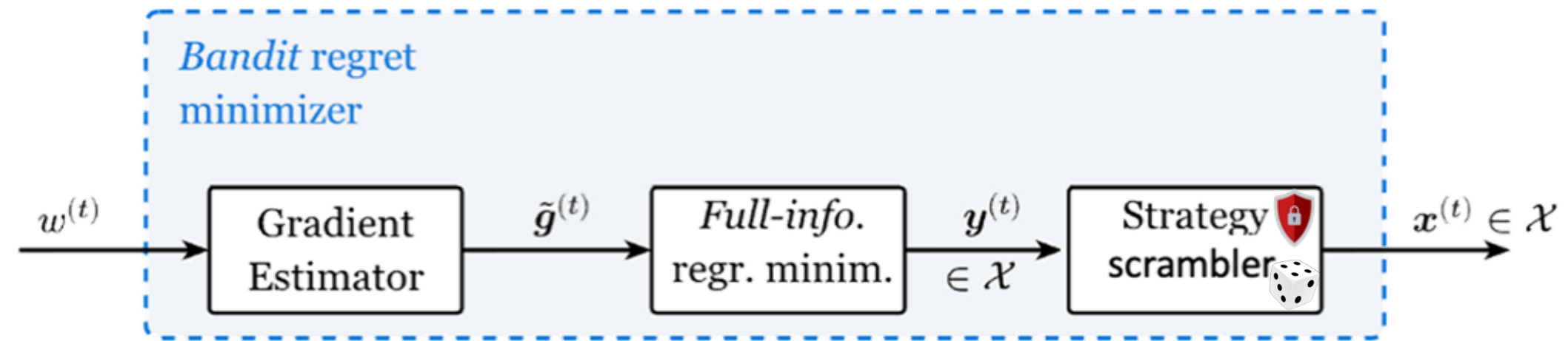
$$\max_{a \in A} \mathbb{E} \left[\sum_{t=1}^T g^{(t)}(a) \right] - \mathbb{E} \left[\sum_{t=1}^T g^{(t)}(a^{(t)}) \right] \leq \frac{T\eta|A|}{2} + \frac{\log |A|}{\eta}.$$

$\mathbb{E}$: w.r.t. randomness in adversary's strategy (if any) and randomness in sampling the $a^{(t)}$'s

Taking $\eta = \sqrt{\frac{2 \log |A|}{|A|T}}$ gives

$$\text{Pseudo-regret} \leq \sqrt{2T|A| \log |A|}.$$

Let's be precise about regret?



Given the randomness, three natural regret guarantees:

$$\text{PseudoReg}^{(T)} := \max_{\hat{\mathbf{x}} \in \mathcal{X}} \mathbb{E} \left[\sum_{t=1}^T \langle \mathbf{g}^{(t)}, \hat{\mathbf{x}} \rangle - \sum_{t=1}^T \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle \right] = o(T)$$

Pseudo-regret

Pseudo-regret $\leq$ Expected regret

$$\mathbb{E} [\text{Reg}^{(T)}] := \mathbb{E} \left[\max_{\hat{\mathbf{x}} \in \mathcal{X}} \sum_{t=1}^T \langle \mathbf{g}^{(t)}, \hat{\mathbf{x}} \rangle - \sum_{t=1}^T \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle \right] = o(T)$$

Expected regret

$$\mathbb{P} \left[\max_{\hat{\mathbf{x}} \in \mathcal{X}} \sum_{t=1}^T \langle \mathbf{g}^{(t)}, \hat{\mathbf{x}} \rangle - \sum_{t=1}^T \langle \mathbf{g}^{(t)}, \mathbf{x}^{(t)} \rangle \leq o(T) \sqrt{\log \frac{1}{\delta}} \right] \geq 1 - \delta$$

High-probability regret